

1. Project context

Bot Hunter is a review-first bot detection pipeline. It keeps the rules layer readable, lets the anomaly model add statistical coverage, and treats all score outputs as unlabeled operational evidence rather than measured ground truth. That matters because the dataset does not include labels, so precision is reported as an operational confidence estimate instead of a calibrated metric.

2. Classifiers

The application implements two classifiers. The first is a rules-based classifier that scores repeated query/domain pairs, repeated queries, high-volume domains, dense region/browser/OS clusters, exact time-to-click reuse, same-second bursts, implausibly fast clicks, moderately long time-to-click values, extreme time-to-click values, and regular pseudo-session inter-arrival timing. Exact time-to-click reuse is selectively calibrated with a 99th-percentile reuse-count cutoff and an absolute floor so the rule can adapt to timer reuse patterns without letting low-count coincidences fire. The second classifier is an Isolation Forest anomaly model over standardized behavioral features. Events that isolate unusually quickly in the fitted forest receive higher anomaly scores.

3. Methods evaluated but not included

Bot Hunter also evaluated HDBSCAN, a shallow autoencoder, and a small VAE as optional anomaly backends in prior benchmark work. They were not promoted because they did not improve reviewer utility enough to justify their runtime, memory, or complexity costs. HDBSCAN was much slower than the current anomaly path on the full dataset, while the neural models used more memory and did not produce clearer explanations than Isolation Forest. The production ML path therefore remains Isolation Forest when scikit-learn is available, with k-means as the lightweight fallback.

4. Anomalies found

The run analyzed 149,239 events and flagged 3,732 events as bots (2.50%). The strongest explainable patterns were:

- repeated query: 3,328 events
- repeated query/domain pair: 2,094 events
- same-second click burst: 1,928 events
- high-volume clicked domain: 1,906 events

- very short query: 1,822 events
- heavy region/browser/os cluster: 1,716 events
- moderately long time-to-click: 546 events
- extreme time-to-click: 276 events
- implausibly fast click: 40 events
- regular inter-arrival timing: 14 events

The dashboard exposes these same signals with sample events so a business user can inspect the likely automated behavior without reading model internals.

5. Filtering options

Practical filters for similar datasets include dropping or quarantining traffic from repeated query/domain pairs, repeated exact ttc values, dense same-second bursts, and events above the combined anomaly threshold. Bot Hunter assigns operational tiers without changing the binary is_bot prediction:

- suppress: 1,073 events
- quarantine: 2,659 events
- monitor: 145,507 events

Use suppress for high-confidence bot traffic after policy approval, quarantine for bot traffic that should be held for review, and monitor for traffic that is not selected for bot action but should remain available for trend analysis and future labels.

6. Method disagreement

The combined score uses a 0.58/0.42 heuristic/ML split because the rules layer is more directly explainable and should remain slightly dominant, while ML still has enough weight to move borderline cases and catch multivariate oddities. The thresholds are conservative guardrails, not learned cutoffs. The same 0.62 heuristic and 0.90 ML agreement thresholds used in suppression are also reported separately so blind spots are visible:

- Heuristic + ML: 1,073 events
- Heuristic only: 3 events
- ML only: 13,851 events
- Neither strong: 134,312 events

7. Threshold rationale

The binary decision uses the stronger of two conservative gates: the event is selected

when its combined score is at or above the run-specific 97.5th-percentile cutoff (0.586199 in this run), or when the rules-only heuristic score reaches 0.62 on its own. The percentile cutoff keeps the submitted bot volume stable for an unlabeled dataset while still letting the anomaly model influence which borderline events enter the review set. The heuristic override prevents high-confidence, explainable rule hits from being missed just because the anomaly ranking moved around after a feature or backend change.

The threshold is not a learned probability boundary. It is an operational cutoff for a review-first workflow where false positives and false negatives are treated as roughly comparable. In this run, the heuristic-only flag rate was 0.72%, while the ML upper-tail reference rate was 1.50%; those rates are reported separately so reviewers can see how much each method contributes before the combined decision is applied.

8. Rationale and generalization

The heuristic model is transparent and easy to convert into policy. Only exact time-to-click reuse uses percentile calibration because it is a global duplicate-count signal whose suspiciousness depends on the dataset's observed timer granularity and reuse distribution; the absolute floor protects against weak duplicate counts in smaller or smoother datasets. Other heuristic cutoffs remain fixed or total-rate based because they represent separate behavioral concepts. The time-to-click timing bands are intentionally tiered: clicks from 0 to 250 ms are treated as implausibly fast direct evidence, clicks from 20 to 60 seconds add low-weight support for delayed or mechanical click patterns, and clicks above 120 seconds remain a separate extreme timing signal. The regular inter-arrival rule is intentionally narrow because the dataset has no explicit user or session identifier: it only compares clicks with the same region, browser, OS, query, and clicked domain, requires at least eight events, and adds low-weight supporting evidence rather than a standalone bot decision. Structured rule contributions include `threshold_mode`, with fixed rules reported as absolute and adaptive exact-ttc reuse reported as `adaptive_percentile` when present. The Isolation Forest model catches multivariate oddities that a small rule set may miss. Both should generalize when bot traffic is repetitive or mechanically timed, but they may miss human-like bots and may over-flag legitimate campaigns that naturally produce high repetition. The thresholds should be recalibrated when traffic mix, geography, or ad inventory changes materially.

9. Probability assessment

The estimated probability that a flagged event is fraudulent is 79%. This is not label-calibrated precision; it is a reasoned estimate based on agreement between independent signals. Events flagged by both the heuristic model and the upper tail of the ML anomaly score are more likely to be fraudulent than events flagged by only one weak signal. The report therefore treats probability as an operational confidence estimate, not a

measured ground truth metric.

10. Known limitations

- The dataset has no ground-truth bot labels, so Bot Hunter cannot report measured precision, recall, or calibration.
- The rules intentionally favor interpretable repetition and timing signals. Human-like automation, slow distributed bot traffic, or attacks spread across many query/domain combinations may be under-detected.
- Legitimate campaigns can create high repetition, dense device clusters, or synchronized bursts. The suppress tier should therefore be policy-approved and periodically sampled.
- The regular inter-arrival rule uses narrow pseudo-session groups because there is no user or session identifier. It is supporting evidence rather than proof of automation.
- The anomaly score is relative to the current traffic mix. Material changes in geography, inventory, campaign volume, browser mix, or feature extraction should trigger a fresh threshold and review calibration.

11. Recommended actions

Assuming false positives and false negatives are roughly equal in cost, the submitted binary prediction uses the combined score threshold rather than only the highest-confidence intersection. For business action, use three tiers: suppress bot events with the strongest combined, heuristic, or heuristic/ML agreement signals; quarantine the remaining bot events for review; and monitor traffic that is not selected for bot action while retaining it for drift checks and future labels.

12. Future work

With more time, I would add labeled validation data, campaign-level normalization, browser/user-agent fingerprinting if available, time-series burst detection by inventory, calibrated probabilities, and a feedback loop from manual review decisions.

- TODO: Add a binary sub-200ms feature.
- TODO: Add click velocity / burst density.
- TODO: Add query entropy.
- TODO: Tune Isolation Forest max_samples and max_features after the new mechanical features land.

13. Submission and decision summary

The repository includes submission.tsv with event_id, is_bot, and operational_tier,

preserving the final binary prediction while adding a workflow tier. This run selected 3,732 of 149,239 events as likely bots (2.50%). The operational split is 1,073 suppress, 2,659 quarantine, and 145,507 monitor events.

Use the binary `is_bot` field as the compatibility output for downstream systems. Use `operational_tier` to decide handling: suppress high-confidence bot traffic after policy approval, quarantine lower-confidence bot traffic for review or delayed action, and monitor the remaining traffic for drift checks and future labels. The report and dashboard should be read as review aids, not as proof of fraud for every individual event.